

SHANTANU DEY

Software Engineer | Java | Spring Boot | REST APIs | Python | Azure Cloud | NoSQL | Maven | Git | Backend Development | API Testing

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SUMMARY

Results-driven Software Engineer with hands-on experience designing and deploying high-throughput RESTful microservices, event-driven architectures, and cloud-native backends. Proficient in Java, Spring Boot, Python, and NoSQL databases, with a proven track record of optimizing database queries, reducing endpoint latency by up to 25%, and integrating secure cloud infrastructure on Microsoft Azure. Strong expertise in API test automation, CI/CD workflows, and scalable distributed system design.

EXPERIENCE

- Software Engineer** 12/2021 – 12/2022
Persistent Systems Ltd. Hinjewadi, Pune
- Architected and tested 15+ core RESTful microservices using **Java** and **Spring Boot**, cutting API latency by **22%** across production calls.
 - Conducted end-to-end API validation with **Postman**, achieving a **99.2%** test-pass reliability rate across Agile sprint releases.
 - Configured and indexed **NoSQL database** schemas, reducing data read latency by **30%** for high-frequency queries.
 - Streamlined builds and package dependencies using **Maven** while managing Git branches within an Agile Scrum framework.
 - Diagnosed and patched 40+ runtime issues and bottlenecks, directly bolstering backend stability and throughput.

PROJECTS

- E-Fuel & E-Charge – Serverless Emergency Logistics**
Microsoft Azure • React • Cosmos DB • Azure Functions • REST APIs
- Engineered an event-driven serverless platform on **Azure** coordinating fuel dispatch and EV charging, reducing emergency response dispatch times by **35%**.
 - Built backend microservices integrating geospatial spatial routing and PESO compliance checks, verifying order handshakes with secure one-time PINs.
 - Configured asynchronous telemetry and cloud workflows using **Azure Functions** and **Cosmos DB**, handling concurrent status updates with sub-200ms latency.
- Enterprise AI Token Management Gateway**
Python • FastAPI • Redis • ClickHouse • Docker • RESTful APIs
- Architected an enterprise API gateway to rate-limit and monitor distributed LLM API consumption across developer teams.
 - Implemented sliding-window rate limiters and token-metering caching algorithms via **Redis**, reducing monthly inference costs by **18%**.
 - Structured high-throughput streaming proxy endpoints in **FastAPI** with persistent audit logging in **ClickHouse**, sustaining **500+ req/sec**.
- Object Detection Using HAAR Cascade & OpenCV**
Python • OpenCV • HAAR Cascade • AdaBoost • Computer Vision
- Built an image recognition solution using **Python** and **OpenCV** to detect and classify target objects across live video feeds and static datasets.
 - Applied **AdaBoost** algorithm to optimize classifier feature selection, achieving an **89% detection accuracy** with minimal compute overhead.
 - Engineered image pre-processing pipelines (grayscale filtering, edge detection, ROI scaling), accelerating real-time video processing to **30+ FPS**.

EDUCATION

- M.Tech Information Security / Cyber Security**
Maulana Abul Kalam Azad University of Technology, Kolkata, India
2024 – 2026
- B.Tech Information Technology**
Techno India Group (BIT), Maulana Abul Kalam Azad University of Technology, Kolkata, India
2016 – 2020

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900)**
Microsoft | May 2022

TRAINING

- Basic Cyber Security Practices**
CDAC – Kolkata | May 2021

SKILLS

- Programming Languages**
Core Java, Python, JavaScript, SQL, HTML, CSS, Bash/Shell Scripting, Golang
- Frameworks & Libraries**
Spring Boot, Hibernate, Node.js, React.js, Angular, RESTful APIs, Bootstrap, jQuery
- Software Architecture**
OOP, Data Structures & Algorithms, Microservices Architecture, Design Patterns, Multithreading, MVC Architecture, Secure Coding
- Databases & Cloud**
MySQL, PostgreSQL, MongoDB, Oracle Database, NoSQL Databases, Azure Cloud, Cosmos DB, Redis
- DevOps & Tools**
Git, GitHub, GitLab, Docker, Kubernetes, Jenkins, CI/CD, Maven, Linux, Postman
- Testing & Practices**
Unit Testing, Integration Testing, JUnit, TestNG, Debugging, Agile Methodology, Scrum
- Cyber Security Tools**
Wireshark, Tcpdump, Nmap, Metasploit Framework, ACUNETIX, Nessus, John the Ripper
- AI Tools & Platforms**
Claude, GitHub Copilot, Replit, OpenAI APIs, Gemini Pro, DeepSeek